A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-22254326})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 1135 & 408 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,1135 + \mathbf{Z}\,(408 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2+22254326 = 0$, of discriminant -89017304. Class group invariants $(170 \quad 10 \quad 5)$, class number 8500; the three stored generator ideals (HNF) are $\begin{pmatrix} 3731 & 33 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 105 & 68 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 274)\, y^3 + (-26s + 124004)\, y^2 + (-160s + 843464)\, y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 549*y^8 + 247460*y^7 + 23768322*y^6 + 1217122232*y^5
+ 38010888968*y^4 + 705937346496*y^3 + 8638102431040*y^2 + 447616934
28224*y + 1113104177521664
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + (-\frac{8277}{1883559343459375})\, s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25722962367664 = 2^4 \cdot 17 \cdot 281 \cdot 336547027$:

$$\begin{pmatrix} 6430740591916 & 152984799903 & 5784278660493 & 2156388354375 & 5388734372544 & 2249715465634 & 3281800058476 & 1874428435981 & 453143422599 & 3552468057593 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 384 factors with signed exponents of absolute value up to 342204713019417024708763852; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 274)\, y^3 + (-26s + 124004)\, y^2 + (-160s + 843464)\, y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 549*y^8 + 247460*y^7 + 23768322*y^6 + 1217122232*y^5
+ 38010888968*y^4 + 705937346496*y^3 + 8638102431040*y^2 + 447616934
28224*y + 1113104177521664
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6102548079616 = 2^{10} \cdot 17 \cdot 761 \cdot 460657$:

$$\begin{pmatrix}
95352313744 & 21665950876 & 22620955972 & 2744162039 & 12983457041 & 40134592562 & 38973344485 & 11396481091 & 82133886882 & 61691623307 \\
0 & 4 & 0 & 1 & 0 & 0 & 0 & 0 & 2 & 2 \\
0 & 0 & 4 & 2 & 2 & 0 & 0 & 3 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 379 factors with signed exponents of absolute value up to 6004437949318878325608650 6; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 274)\, y^3 + (-26s + 124004)\, y^2 + (-160s + 843464)\, y + (-7040s - 3184992)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 549*y^8 + 247460*y^7 + 23768322*y^6 + 1217122232*y^5
+ 38010888968*y^4 + 705937346496*y^3 + 8638102431040*y^2 + 447616934
28224*y + 1113104177521664
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $623352706772 = 2^2 \cdot 757 \cdot 205862849$:

$$\begin{pmatrix}
311676353386 & 217593510310 & 233217344991 & 89392999167 & 175814013287 & 155607433997 & 5837790543 & 124241934306 & 291997499085 & 267081267908 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 385 factors with signed exponents of absolute value up to 681695366085403329350887838; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1307)\, y^3 + (14s - 45179)\, y^2 + (-323s + 311148)\, y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2615*y^8 - 92972*y^7 + 24675229*y^6 - 772853530*y^5 +
21623637605*y^4 - 76125109920*y^3 + 1106615447646*y^2 + 52945002665
384*y + 662321427559016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right) s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $144573632899832 = 2^3 \cdot 211 \cdot 419 \cdot 1013 \cdot 201787$:

$$\begin{pmatrix}
36143408224958 & 9507047005235 & 7620525053315 & 21332342536837 & 435178638397 & 33744057194394 & 9902379060764 & 34843942065700 & 23616304529659 & 1087038676835 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 373 factors with signed exponents of absolute value up to 109536118334095704163433634; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1307)\, y^3 + (14s - 45179)\, y^2 + (-323s + 311148)\, y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2615*y^8 - 92972*y^7 + 24675229*y^6 - 772853530*y^5 +
21623637605*y^4 - 76125109920*y^3 + 1106615447646*y^2 + 52945002665
384*y + 662321427559016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1827774298048 = 2^6 \cdot 41 \cdot 293 \cdot 2377339$:

$$\begin{pmatrix} 57117946814 & 15180445462 & 31956016916 & 38387689546 & 15130530567 & 6607672904 & 15488499894 & 42971109092 & 7000575850 & 41391073478 \\ 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 373 factors with signed exponents of absolute value up to 96493234780934689694303003; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s + 1307)\, y^3 + (14s - 45179)\, y^2 + (-323s + 311148)\, y + (-4354s - 15506100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 2615*y^8 - 92972*y^7 + 24675229*y^6 - 772853530*y^5 +
21623637605*y^4 - 76125109920*y^3 + 1106615447646*y^2 + 52945002665
384*y + 662321427559016
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5451308556536 = 2^3 \cdot 17 \cdot 211 \cdot 521 \cdot 364621$:

$$\begin{pmatrix} 1362827139134 & 334216782147 & 483274228601 & 1124039577423 & 1136764613279 & 463585673214 & 1296261141754 & 1317876116578 & 111497716197 & 275895654537 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 368 factors with signed exponents of absolute value up to 6745824587221871426189 6352; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)\, y^2 + (-60s - 2693904)\, y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6823*y^8 + 83156*y^7 + 6177604*y^6 + 65718384*y^5 + 2
1744501692*y^4 - 1273411417248*y^3 + 7865269518176*y^2 - 93442608864
3840*y + 41140723099404800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + (-\frac{8277}{1883559343459375})\, s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $49757027237108 = 2^2 \cdot 17 \cdot 731720988781$:

$$\begin{pmatrix} 24878513618554 & 2611879157078 & 4402196056015 & 15171208038109 & 10707226067999 & 22182452292499 & 15277990837132 & 10840657748104 & 17129997095411 & 13541848265407 \\ 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 306 factors with signed exponents of absolute value up to 2266523851800; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)\, y^2 + (-60s - 2693904)\, y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6823*y^8 + 83156*y^7 + 6177604*y^6 + 65718384*y^5 + 2
1744501692*y^4 - 1273411417248*y^3 + 7865269518176*y^2 - 93442608864
3840*y + 41140723099404800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + (\frac{4169}{10356949124803944})\, s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55895963497600 = 2^7 \cdot 5^2 \cdot 17^3 \cdot 3555361$:

$$\begin{pmatrix} 6044113700 & 4827625660 & 5703618966 & 5116146198 & 4149037835 & 2328605976 & 4511401936 & 5605107662 & 4713555080 & 1182972825 \\ 0 & 34 & 0 & 27 & 20 & 0 & 27 & 11 & 8 & 8 \\ 0 & 0 & 34 & 22 & 1 & 2 & 26 & 23 & 25 & 9 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 299 factors with signed exponents of absolute value up to 2086164805237; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 3412y^3 + (-10s + 38166)\,y^2 + (-60s - 2693904)\,y + (26320s + 160387680)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 6823*y^8 + 83156*y^7 + 6177604*y^6 + 65718384*y^5 + 2
1744501692*y^4 - 1273411417248*y^3 + 7865269518176*y^2 - 93442608864
3840*y + 41140723099404800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10479774113920 = 2^7 \cdot 5 \cdot 113 \cdot 144908381$:

$$\begin{pmatrix}
327492941060 & 20730696176 & 213806419162 & 243180723120 & 290624740070 & 88676904391 & 1097384690 & 148328917835 & 272459835199 & 37327355968 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 305 factors with signed exponents of absolute value up to 3374984441326; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6896*y^8 + 16390648*y^6 + 35751264192*y^4 + 15219120431376*y^
2 + 502536151586304
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + \left(-\frac{8277}{1883559343459375}\right) s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $65730985358445 = 3^2 \cdot 5 \cdot 11 \cdot 73 \cdot 1819039307$:

$$\begin{pmatrix}
21910328452815 & 21443205004476 & 18305065303343 & 15108083457521 & 9087095186500 & 4367738330771 & 3448605092181 & 3966745825383 & 2991267132982 & 14858222736136 \\
0 & 3 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 118 factors with signed exponents of absolute value up to 10049343475; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6896*y^8 + 16390648*y^6 + 35751264192*y^4 + 15219120431376*y^
2 + 502536151586304
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $201757417959375 = 3^2 \cdot 5^5 \cdot 17 \cdot 241 \cdot 421 \cdot 4159$:

$$\begin{pmatrix} 1614059343675 & 1482987640910 & 710592456315 & 1583357826880 & 1025715615623 & 381656838563 & 238205790518 & 1063541738071 & 571177889553 & 815939166601 \\ 0 & 5 & 0 & 0 & 0 & 2 & 2 & 2 & 0 & 4 \\ 0 & 0 & 5 & 0 & 2 & 0 & 1 & 4 & 4 & 0 \\ 0 & 0 & 0 & 5 & 2 & 3 & 4 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 11911056261; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3448y^3 + 48sy^2 + 2250972y + 4752s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6896*y^8 + 16390648*y^6 + 35751264192*y^4 + 15219120431376*y^
2 + 502536151586304
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1683124114212 = 2^2 \cdot 3^9 \cdot 17 \cdot 67 \cdot 137^2$:

$$\begin{pmatrix}
8426322 & 3218541 & 6901113 & 6821134 & 2201875 & 5259342 & 760 & 3046288 & 3048563 & 5851911 \\
0 & 411 & 354 & 274 & 132 & 149 & 272 & 71 & 232 & 23 \\
0 & 0 & 9 & 3 & 8 & 8 & 3 & 5 & 6 & 8 \\
0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 2 & 1 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 11758747692; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 2831)\, y^3 + (-7s + 216964)\, y^2 + (250s + 2567748)\, y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 5678*y^8 + 411280*y^7 + 33668671*y^6 + 1563064988*y^5
+ 51400880486*y^4 + 1144437592296*y^3 + 17335862947848*y^2 + 115771
902139520*y + 477791523455936
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + (-\frac{8277}{1883559343459375})\, s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $715380577792 = 2^9 \cdot 31^3 \cdot 46901$:

$$\begin{pmatrix}
5815724 & 3898002 & 4407890 & 1678234 & 3070360 & 900940 & 1139524 & 3303839 & 5012874 & 5044946 \\
0 & 62 & 0 & 28 & 40 & 34 & 36 & 43 & 20 & 26 \\
0 & 0 & 62 & 48 & 36 & 8 & 6 & 45 & 18 & 39 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 11518664377183; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 2831)\, y^3 + (-7s + 216964)\, y^2 + (250s + 2567748)\, y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 5678*y^8 + 411280*y^7 + 33668671*y^6 + 1563064988*y^5
+ 51400880486*y^4 + 1144437592296*y^3 + 17335862947848*y^2 + 115771
902139520*y + 477791523455936
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + \left(\frac{4169}{10356949124803944}\right) s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $151644131083504 = 2^4 \cdot 17 \cdot 31 \cdot 311 \cdot 57827527$:

$$\begin{pmatrix}
37911032770876 & 35495499484903 & 13356946055389 & 19446838963646 & 17133079054323 & 6032181862718 & 26102857327172 & 7322424794045 & 440213513907 & 28451441224843 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 12920316257821; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 4y^4 + (-s + 2831)\, y^3 + (-7s + 216964)\, y^2 + (250s + 2567748)\, y + (344s + 21798120)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8*y^9 + 5678*y^8 + 411280*y^7 + 33668671*y^6 + 1563064988*y^5
+ 51400880486*y^4 + 1144437592296*y^3 + 17335862947848*y^2 + 115771
902139520*y + 477791523455936
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $14072221709552 = 2^4 \cdot 31^2 \cdot 167 \cdot 2341^2$:

$$\begin{pmatrix}
24238714 & 13861061 & 18923111 & 17119795 & 23318803 & 22599051 & 9729395 & 17491024 & 4606488 & 2240226 \\
0 & 72571 & 26282 & 59587 & 11705 & 43573 & 43360 & 16012 & 46367 & 11926 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 317 factors with signed exponents of absolute value up to 15969045174199; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 352*y^8 - 1228674*y^6 + 7432100024*y^4 - 425306254511*y^2 + 2
3426594876376
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18945811}{1883559343459375} + (-\frac{8277}{1883559343459375})\, s, \qquad J = \begin{pmatrix} 1135 & 408 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1135$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4042709872992 = 2^5 \cdot 3^5 \cdot 7 \cdot 313 \cdot 237287$:

$$\begin{pmatrix}
9358124706 & 7253214804 & 758904330 & 1488470346 & 4488292088 & 2993093001 & 8388389922 & 8413685229 & 3439242581 & 4416695629 \\
0 & 6 & 0 & 0 & 2 & 2 & 0 & 0 & 1 & 0 \\
0 & 0 & 6 & 0 & 4 & 0 & 2 & 3 & 2 & 1 \\
0 & 0 & 0 & 6 & 4 & 4 & 3 & 1 & 1 & 3 \\
0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 292647140; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 352*y^8 - 1228674*y^6 + 7432100024*y^4 - 425306254511*y^2 + 2
3426594876376
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{23465105}{5178474562401972} + (\frac{4169}{10356949124803944})\, s, \qquad J = \begin{pmatrix} 2106 & 982 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2106$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $132782106393216 = 2^7 \cdot 3^2 \cdot 7^4 \cdot 29 \cdot 1655377$:

$$\begin{pmatrix} 395184840456 & 214112868564 & 2290539928 & 125438298108 & 283117162830 & 373445348788 & 77019394157 & 241016247860 & 91004307413 & 120539806415 \\ 0 & 21 & 14 & 15 & 8 & 18 & 7 & 18 & 10 & 3 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 8 & 7 & 4 & 7 & 4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 3011982; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 176y^3 + 18sy^2 - 629825y - 1026s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 352*y^8 - 1228674*y^6 + 7432100024*y^4 - 425306254511*y^2 + 2
3426594876376
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{308187161}{134426863868415625} + \left(-\frac{42102}{134426863868415625}\right) s, \qquad J = \begin{pmatrix} 2665 & 1033 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2665$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $23853906432 = 2^9 \cdot 3^{11} \cdot 263$:

$$\begin{pmatrix} 511272 & 59862 & 235812 & 414250 & 65256 & 316837 & 142624 & 164558 & 74929 & 323076 \\ 0 & 6 & 0 & 2 & 0 & 2 & 4 & 4 & 5 & 0 \\ 0 & 0 & 12 & 6 & 6 & 10 & 2 & 2 & 6 & 6 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 54 & 9 & 46 & 46 & 18 & 46 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 72831596; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.